CSS Grid

- Allows layout for child elements
- In vertical and horizontal grid
- Blocks can "span" multiple "cells"
- We used "areas"
    - With different column sizes
    - Not the only option

CSS Grid Properties

We will be using some additional CSS properties

Parent element: (with `display: grid;`)

- `grid-template-columns: repeat(12, 1fr);`
    - Creates even distribution of space in columns
    - `12` can be any number
- `gap` - puts space between columns/rows

CSS Properties Child Element

- `grid-column: span 1;`
    - The `1` can be any number
    - Child will span that many cells
    - Spans over any gap
- `grid-column: 3 / 6;`
    - The `3` and `6` can be any number
    - Think of the `/` as "to"
    - Child will span from "line" 3 to "line" 6
    - Lines are the start/end of cell boundaries
    - Line 1 is left side of first cell
    - Line 2 is right of first cell, left of 2nd

Not Areas

- We used `grid-template-areas` and `grid-area`
    - Those are still valid
    - But we aren't using them here
    - Different way of laying cells out

Grid layout without display: grid

- Grid column layout is a design idea
    - CSS Grids are a good way to implement
    - ...but not the only way
    - Can do with flex
    - Can do with percentage widths
- Concept vs implementation
    - Different
    - ...but please use css grid

display: grid impacts direct children

Remember how `display: grid;` works

- Only affects direct children
- To match non-child elements to grid
    - Make wrapper width 100%
    - Recreate the same grid inside wrapper

Grid column layouts

A common design pattern

- Page layout aligns to evenly spaced columns
- A row will contain items
    - Spanning cells
    - Evenly or unevenly

Various Online Grid Layouts - Good and Bad

Most not using CSS Grids, but are multi column grids

- **https://wines.com/**
- **https://bostonglobe.com/**
- **https://bbc.com/**

Compare these:

- **https://allrecipes.com/** - consistent "break out"
- **https://www.usfigureskating.org/** - inconsistent

"Choppy" - Better if they used a grid:

- **https://www.usarchery.org/**

Previous Iteration of NEU website

https://web.archive.org/web/20221231134920/https://www.no

- → Admissions
- → Visit Campus
- → Admissions
- → PhD Education
- → Security
- → Sustainability
- → Campus Network
- → Admissions



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TRENDING EVENTS

Nov 17
GSS Coffee & Conversations: Being in

Nov 17
DH Office Hours: GIS Day Celebration

Nov 18
Space Diplomacy workshop

Nov 18
In Pursuit of Knowledge: Black Women and

Responsive Alignment

Benefits of this column layout

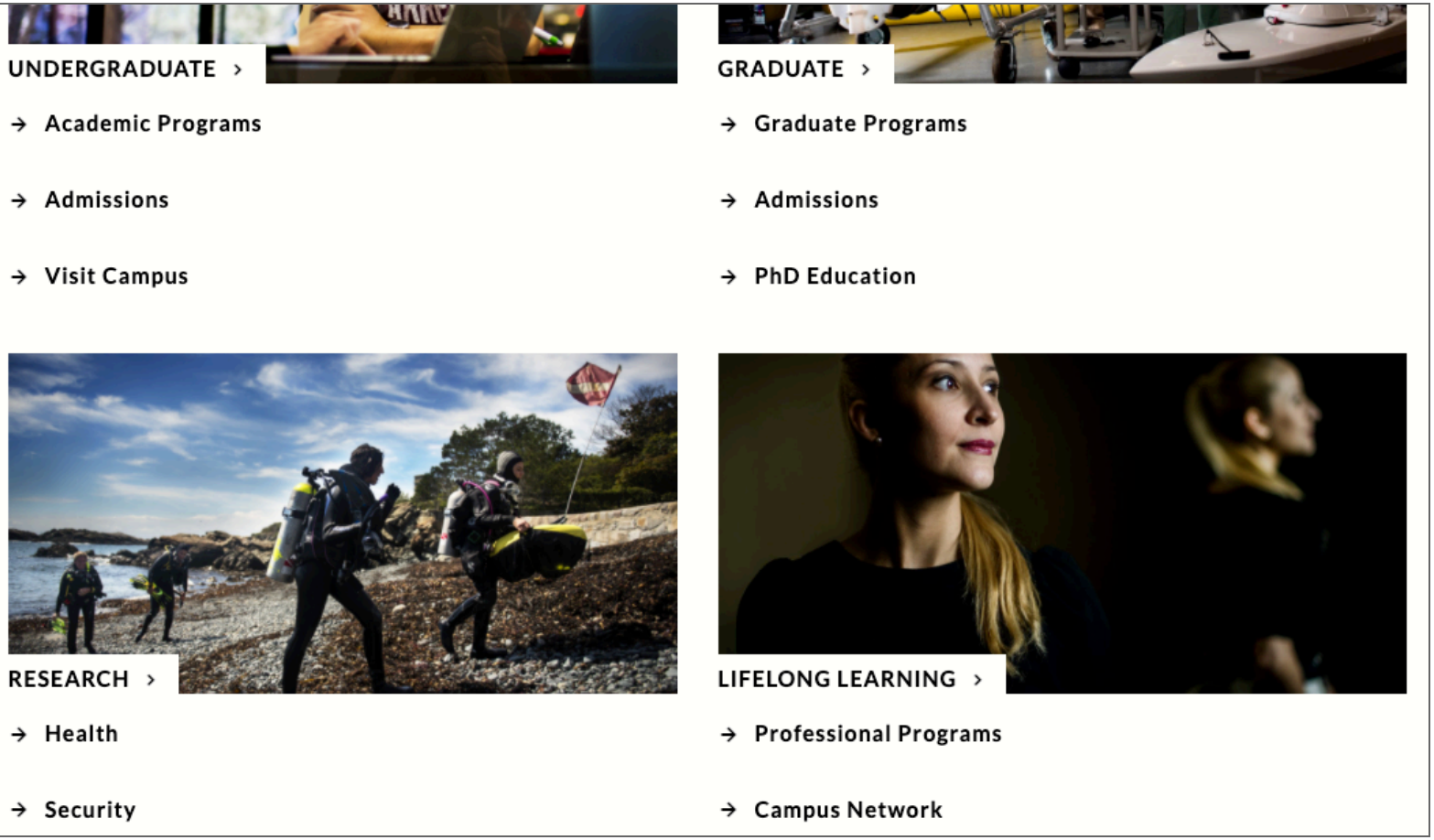
- Allows mix of sizes
- Consistent alignment
    - Not off by a pixel or two
- Don't need to declare areas
    - "rows" can auto wrap
- Easily responsive
    - Cells evenly expand

Adaptive Alignment

Another potential benefit

- Change the cells elements span
    - At different breakpoints
        - Mobile/Tablet/Desktop?

Adaptive Tablet Example



Adaptive Phone Landscape Example

RESEARCH ›

→ Health

→ Security

→ Sustainability

LIFELONG LEARNING ›

→ Professional Programs

→ Campus Network

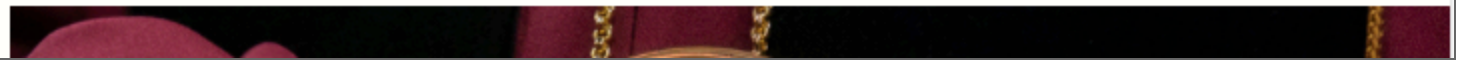
→ Admissions



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12 Column Grid

- Many grid layouts exist
- 12 column is very common
    - Allows options for even spanning:
        - 6 columns, 4 columns, 3 columns
    - Allows offset combos:
        - 3+9, 4+8

Why 12 columns though?

- Good number of options
    - Mostly
- All subjective

Rule of Thirds

In photography, the **Rule of Thirds** says

- Divide picture up into thirds
- Vertically and horizontally
- The 4 intersections are more "visually interesting"

https://en.wikipedia.org/wiki/Rule_of_thirds

12 Column Format allows for thirds

Web Design is a little different than photography

- Photos don't scroll

But you can still use that concept

- A 3+6 or 6+3 split
    - Puts center of an item at "interesting" point
- A 4+8 or 8+4 can work if edges are interesting

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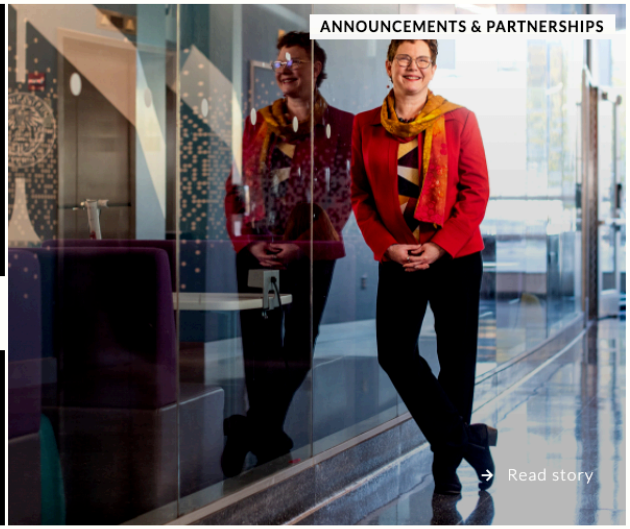
Less than half of US college students know their schools' vaccination policies, study shows ›

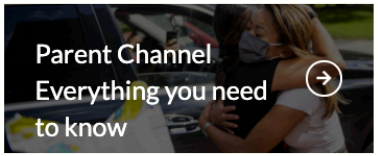
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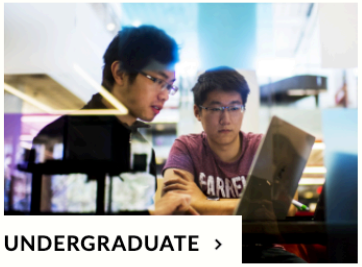
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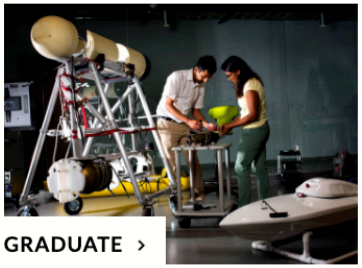
MEDIA INQUIRIES



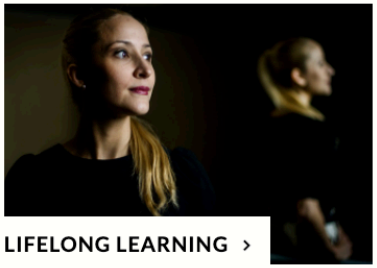





UNDERGRADUATE ›


GRADUATE ›


RESEARCH ›


LIFELONG LEARNING ›

Summary - CSS Grids

- We already used `grid-template-areas`
- For multi-column layout
    - Can use `grid-template-columns`
        - With `repeat()`
        - For evenly sized columns
        - Optionally `gap`
- `grid-column`
    - Can take `span` to span cells
    - Can use `/` to range from/to boundaries

Summary - 12 column grid

- Many grid layouts exist
    - Can be done without CSS Grids
        - But why?
- 12 column very common in web dev
    - Allows for different splits
- Grids are naturally responsive

Summary - Adaptive Grids

Grids can make it easy to have adaptive breakpoints

- Change the number of total columns
    - Items will "wrap" to different rows
- Change the span of different items
    - Will be a reasonable size

You can have multiple adaptive breakpoints